

Week 6

Design & Democracy

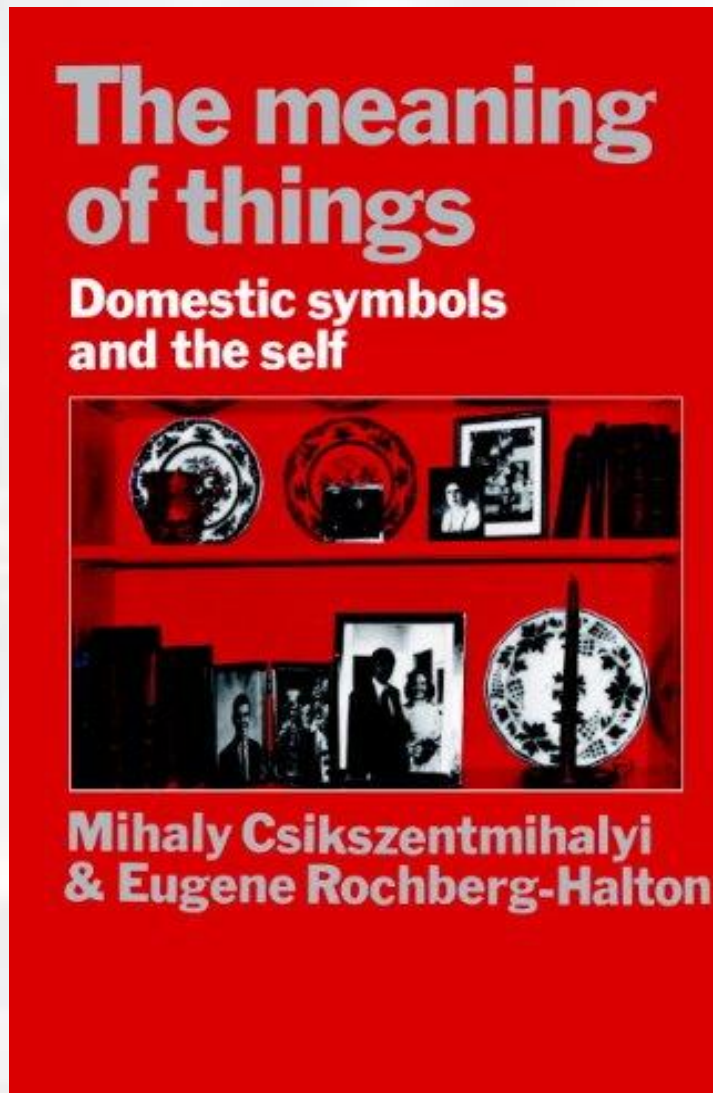
**Social Theory &
Design Thinking (2410302)**

PGS, Fall 2025

Keerati Chenpitayaton

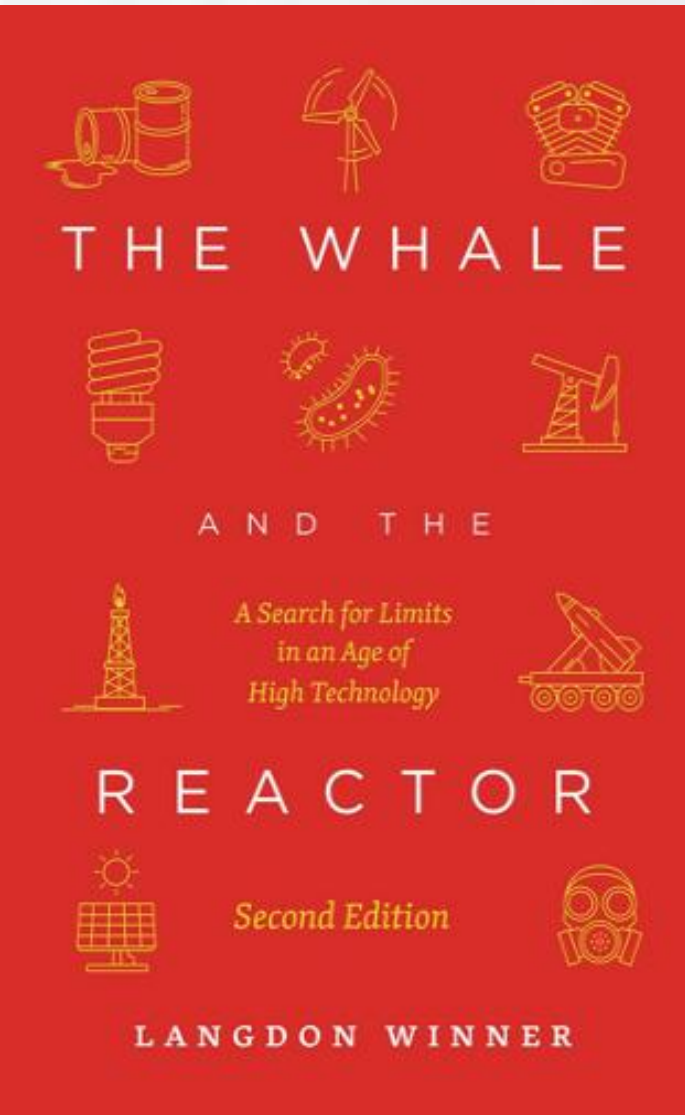
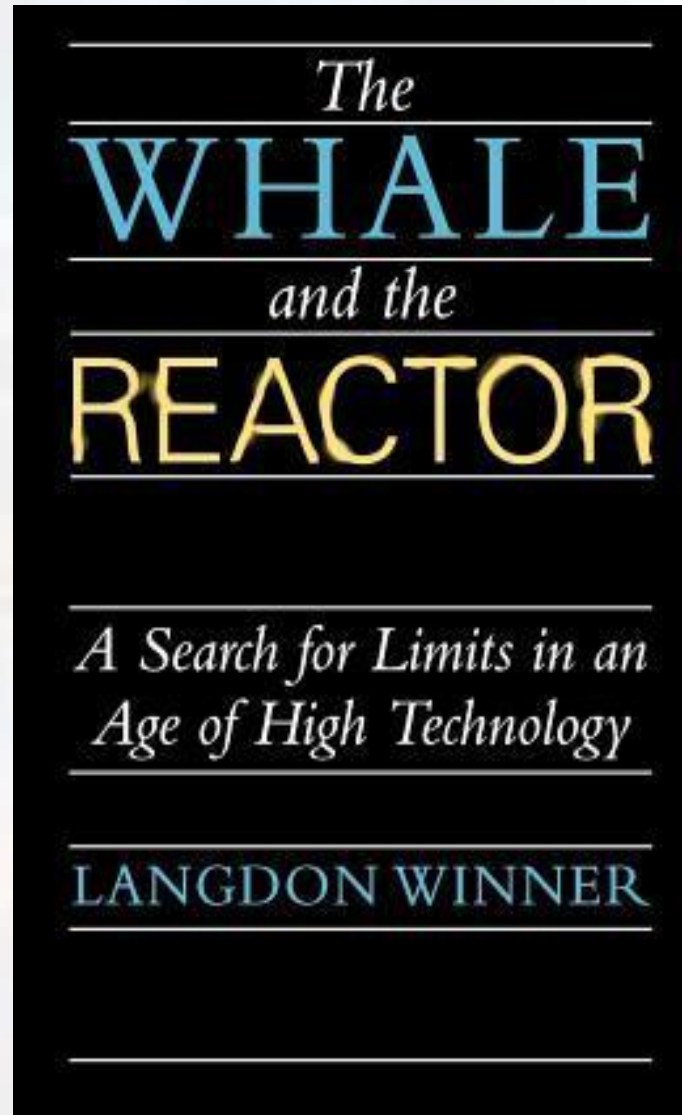
Last Week's Content (Week 5)

- **Design & Self**
- **Mihaly Csikszentmihalyi & Eugene Rochberg-Halton (C&R):** *The Meaning of Things: Domestic Symbols and the Self* (1981)
- **Sherry Turkle:** *Evocative Objects: Things We Think With* (2007)



Assigned Readings

- Langdon Winner, *The Whale and the Reactor: A Search for Limits in the Age of High Technology* (1986/2020)
- **Required Readings:** 2nd Edition: "Part I: A Philosophy of Technology": "Chapter 1: Technologies as Forms of Life," pp. 1-18; "Chapter 2: Do Artifacts Have Politics?," pp. 19-39; and "Chapter 3: Techne and Politeia," pp. 40-58.
- **Recommended Reading:** 2nd Edition: "Chapter 11: Beyond Techno-Narcissism: Self and Other in the Digital Public Realm," pp. 179-196. [Skim the Entire Book]





Chapter 1: Technologies as Forms of Life

Technologies as Forms of Life (1)

- Winner claims that technology is important to social life. We all seem to admit it. Then, what went wrong?
- What is the problem for Winner?
- If it is so important why hasn't there been enough critical reflections upon it in the past.
- "It is reasonable to suppose that a society thoroughly committed to making artificial realities [technology] would have given a real deal of thought to the nature of that commitment" (p. 3).

Technologies as Forms of Life (2)

- **What is a philosophy of technology?**
- “The basic task for a philosophy of technology is to examine critically the nature and significance of artificial aids to human activity” (p. 4).
- We need more of such critical reflection of technology in our societies, in which technologies are becoming part of 21st century social landscapes.
- **Big Questions: “Why is it that the philosophy of technology has never really gotten under way?”**
- “Why has a culture so firmly based upon countless sophisticated instruments, techniques, and systems remained so steadfast in its reluctance to examine its own foundations?” (p. 5)

Technologies as Forms of Life (3)

- Let's in the opposite direction of engineering? (Applied Science of Design or Construction)
- The [American Engineers' Council for Professional Development](#) (ECPD, the predecessor of [ABET](#))^[1] has defined "engineering" as:
- "[T]he creative application of scientific principles to design or develop structures, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behavior under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property."
- Engineers are still *uncritical* about technology when, in fact, their profession has to deal a lot with technology and artificial things.

Technologies as Forms of Life (4)

- **Two Reasons:** 1) The Idea of Progress (i.e., Blaming the 19th Century Social Thought) [Technology → Progress; Technology = Tools for Achieving Progress] and 2) The Separation of the Categories of “Making” and “Use”
- The conventional idea of what technology is and what it means, an idea powerfully reinforced by familiar terms used in everyday language, needs to be overcome if a critical philosophy of technology is to move ahead.

Technologies as Forms of Life (5)

- **The crucial weakness of the conventional idea is that it disregards the many ways in which technologies provide structure for human activity.**
- “Since, according to accepted wisdom, patterns that take shape in the sphere of ‘making’ are of interest to practitioners alone, and since the very essence of ‘use’ is its occasional, innocuous, nonstructuring occurrence any further questioning seems irrelevant.” (p. 6).

Technologies as Forms of Life (6)

- **Neutrality vs. Non-Neutrality of Technology THESIS**
- Conventional Thinking (Something You Simply Take for Granted or Passively Accept without Questions or Skepticisms) = It's Just There! (Technology as a Permanent Feature in Our Society. It's natural, fixed, and static and seems to exist with its own autonomous force.)
- Thinking Against Convention (NOT to Take Things You See or Hear for Granted or Passively Accept without Any Confrontations with It) = It Doesn't Have to Be There. (Technology Is NOT a Permanent Feature of Our Society.)

Technologies as Forms of Life (7)

- Neutrality vs. Non-Neutrality of Technology THESIS
- Conventional Thinking = **Technology Is “Neutral”** → as External Objects with Autonomous Forces, Just There!!!
- We’re the Users. Techs Don’t Have Meanings or Impacts without Us. They’re Neutral.
- Thinking Against Convention = **Technology Has More Implications. It Is Embedded in Our “Forms of Life” and “Distinctive Worlds.”**

Technologies as Forms of Life (8)

- **Technological Determinism vs. Technological Somnambulism**
- **Technological Somnambulism** is a concept used when talking about the philosophy of technology.
- Winner puts forth the idea that we are simply in a state of *sleepwalking* in our mediations with technology.
- This *sleepwalking* is caused by a number of factors.

Technologies as Forms of Life (9)

- One of the primary causes is the way we view technology as tools, something that can be put down and picked up again [Progress → Techs as Tools Helping Humans Progressing]. Because of this view of objects [techs] as something we can easily separate ourselves from, and we fail to look at the long term implications of using that object [tech].
- A second factor is separation of those who make the technology and those who use the technology. This division causes there to be little thought and research going into the effects of using/developing that technology.
- The third and most important idea is the way in which technology seems to create new worlds in which we live ["distinctive worlds" and "forms of life"]. These worlds are created by the restructuring of the common and seemingly everyday things around us. In most situations the changes take place with little attention or care from us because we are more focused on the menial aspects of the technology. **[This is Winner's Counterclaim against the First Two Factors Giving Rise to Technological Somnambulism.]**

Technologies as Forms of Life (10)

- **Technological Determinism: Technology as Cause, Everything Else in Society as Effect**
- Instrumental/Functional ("Effectiveness," "Efficiency," "Productivity," "Impact," "Risk," etc.) **vs. Philosophical (Critical Reflection)**
Understanding of Technology that Winner is trying to put forward.
- Technological Assessment (Simple and Straightforward Cause-and-Effect Analysis): [Technology as Cause] → Impacts on Society as Effect]
- Note: In philosophy, technological determinism is opposed to social constructionism/constructivism. Winner's position seems to be in between the two poles, while leaning toward constructivism.

Technologies as Forms of Life (11)

- **Technology → Use → “Forms of Life” [Ludwig Wittgenstein] (p. 11)**
- Through our use, technology becomes woven into the **texture of our life**. For Examples: Television, the Internet, Home Labor-Saving Devices, etc.
- “It is important to ask, where, if at all, have modern technologies added fundamentally new activities to the range of things human beings do? Where and how have innovations in science and technology begun to alter the very conditions of life instead?” (p. 13)

Technologies as Forms of Life (12)

- Mostly technological innovations were the transformations of old patterns.
- Rarely did they were truly revolutionary in the sense that they transformed the fundamentals of social life.
- **Marx:** Material Conditions of Production and Social Life → Capitalist Mode of Production Limiting the Human Nature and Potential
- **Philosophy:** People Coming together to Create Their World, Their Fabrics of Social Life
- “As we ‘make things work,’ what kind of world are we making?” (p. 17)



Chapter 2: Do Artifacts Have Politics?

Do Artifacts Have Politics? (1)

- **Technological things have political qualities.**
→ Technological things embody specific forms of power and authority. (p. 19)
- Social Determination of Technology (or Social Determinism) → Technology itself does not have political qualities but the socioeconomic system it is embedded in has. [Winner does not aim to go that far.]
- Doing Detective Works to Reveal the Power Holders/Relations behind the Creation of New Technologies
- Technological Politics (p. 21) → Paying Attention to the Intrinsic Characteristics and Meanings of Technological Things [in lieu of Reducing Everything to Social Forces] (p. 22)

Do Artifacts Have Politics? (2)

- **2 Ways in which Artifacts Can Contain Political Properties:**
- **ONE**: Instances in which the invention, design, or arrangement of a specific technical device or system becomes a way of settling an issue in the affairs of a particular community [Seen in the proper light, examples of this kind are fairly straightforward and easily understood.] (p. 22)
- **TWO**: Some cases of what can be called 'inherently political technologies,' man-made systems that appear to require or to be strongly compatible with particular kinds of political relationships [Arguments about cases of this kind are much more troublesome and closer to the heart of the matter.] (p. 22)

Do Artifacts Have Politics? (3)

- With regard to the first way [ONE], two kinds of choices that can affect the relative distribution of power, authority, and privileges in a community. (p. 27)
- #1: Often the crucial decisions is a simple 'yes' or 'no' choice—are we going to develop and adopt the thing or not? (p. 27)
- #2: A second range of choices, equally critical in many instances, has to do with specific features in the design or arrangement of a technical system after the decision to go ahead with it has already been made. (p. 28)
- Great Conclusion: “The things we call ‘technologies’ are ways of building order in our world.” (pp. 28-29)

Do Artifacts Have Politics? (4)

- According to the second way [TWO]—the belief that some technologies are by their very nature political in a specific way—the adoption of a given technical system unavoidably brings with it conditions for human relationships that have a distinctive political cast—for example, centralized or decentralized, egalitarian or inegalitarian, repressive or liberating.
- Authoritarian vs. Democratic Traditions of Technology
- “Certain kinds of technology do not allow flexible designs or arrangements. To choose them is to choose unalterably a particular form of political life.” (p. 29)
- Not every large form of technology (“sociotechnical system”) can be run democratically.

Do Artifacts Have Politics? (5)

- Material (Technological) Basis → Societal Adaptation to Its Sophistication (p. 31) [Technological Determinism; Winner wants to steer clear of this extreme pole.]
- What, after all, does modern technology *make possible or necessary* in political life? (p. 32)

Do Artifacts Have Politics? (6)

- Two Versions of Claim that *Technological Things Are Inherently Political* (p. 32):
- #1: The adoption of a given technical system actually requires the creation and maintenance of a particular set of social conditions as the operating environment of that system. [Techno-Scientific-Industrial-Military Elite] → *Internal (Inherent) Conditions*
- For Examples: Coal, Oil, and Nuclear Power, Cottons and Railways → *Internally* Requiring Particular Sets of Social Conditions to Run

Do Artifacts Have Politics? (7)

- #2: A given kind of technology is strongly compatible with, but does not strictly require, social and political relationships of a particular stripe. → *External (Imposed)*
- For Example: Solar Energy System (Compatible with Democratic Social Organizations) → *Externally Imposed* into the Sociotechnical System
- *External (Imposed) vs. Internal (Inherent) Conditions of Sociotechnical System (p. 33)*
- Other Examples: Atomic Bombs and Traditional Business Enterprises



Chapter 3: *Technē and Politeia*

Technē and Politeia (1)

- All forms of technology and their corresponding forms of social life must be examined to see whether they are *friendly or unfriendly* to the idea of a just society.
- What does the study of politics have to contribute to our thinking about the realm of instrumental things? Where can one turn to look for a political theory of technology? (p. 40)
- Analogy Rooted in Western Political Thought: Practice of Technology → Political Practice [One-Sided Metaphor → Mechanical Metaphor to Illuminate the Art of Constitution Making and Statecraft]

Technē and Politeia (2)

- **Plato:** Statecraft = *technē* = One of the practical arts = good constitutions = well-crafted products of political architecture [Political affairs should not be left to amateurs or the democratic masses.]
- **Plato:** Technology (*technē*) serves as a model for politics and not the other way around. It shows his discomfort and suspicious attitude toward technology until these days. [Technological Foundation of Politics.]
- However, most people who have relied on this attitude have still understood technology in a narrow focused sense.

Technē and Politeia (3)

- “Most of them have politely ignored the substance of technical life, hoping perhaps that it would remain segregated in a narrowly defined corner of human life. Evidently, it did not occur to anyone that Plato’s pregnant analogy would at some point qualify in reverse, that technē itself might become politeia, that technological forms of life might in themselves play a powerful role in shaping society” (p. 41). [Politics serves as a model or foundation of technology → Political Foundation of Technology]
- **Rousseau:** One-Sided Metaphor → Mechanical Metaphor to Illuminate the Art of Constitution-Making [Lawgiver = Architect]
- **The Federalist Papers:** Science of Politics

Technē and Politeia (4)

- Wealth/Material Progress → Vices and Corruption
- Civic virtue and material prosperity are antithetical.
- “The indolent, pleasure-seeking habits of luxurious living tend to subvert qualities of frugality, self-restraint, and self-sacrifice needed to maintain a free society. By implication, any society that wishes to maintain civic virtue ought to approach technical innovation and economic growth with the utmost caution” (p. 43).

Technē and Politeia (5)

- American Exceptionalism (The Progressive Era) → A Fascination with *Efficiency* (p. 46)
- Some distinctive pattern of modern industrial production has risen. → the Constitution of a Sociotechnical Order → The Way of Arranging People and Things (p. 47)
- From a contemporary vantage point, the characteristics of this sociotechnical order embody answers to age-old political questions (5 Characteristics; pp. 47-48).

Technē and Politeia (6)

- 5 Characteristics (pp. 47-48)

Technē and Politeia (7)

- Political Questions (p. 49) → have until now been answered in terms of instrumental language of *efficiency* and *productivity*. (p. 53)
- SOLUTION: Can our society establish forms and limits for technological change, forms and limits that derive from a positively articulated idea of what society ought to be?
[This is the (sub)title of Winner's book!]
- What is the best form of political society? What forms of technology are compatible with the kind of society we want to build?

Technē and Politeia (8)

- **Regimes of Instrumentality (pp. 54-55):** When *technē* becomes *politeia*
- “The idea that society might try to guide its sociotechnical development according to self-conscious, critically evaluated standards of form and limit can no longer be considered a ‘heroic decision’; it is simply good sense. Because technological innovation is inextricably linked to processes of social reconstruction, any society that hopes to control its own structural evolution must confront each significant set of technological possibilities with scrupulous care” (p. 54).

Technē and Politeia (9)

- “The important task becomes, therefore, not that of studying the ‘effects’ and ‘impacts’ of technical change, but one of evaluating the material and social infrastructures specific technologies create for our life’s activity. We should try to imagine and seek to build technical regimes compatible with freedom, social justice, and other key political ends” (p. 55).
- **A process of technological change disciplined by the political wisdom of democracy (p. 55) → Democratic Decision Making (p. 56) →** “must built into the process as early as possible from the beginning otherwise problems will be hard to fix” (p. 57).

Next Week's Theme:

Week 7 + Made-Up Class: Design & Gender & Sexuality

